

Experiment 8

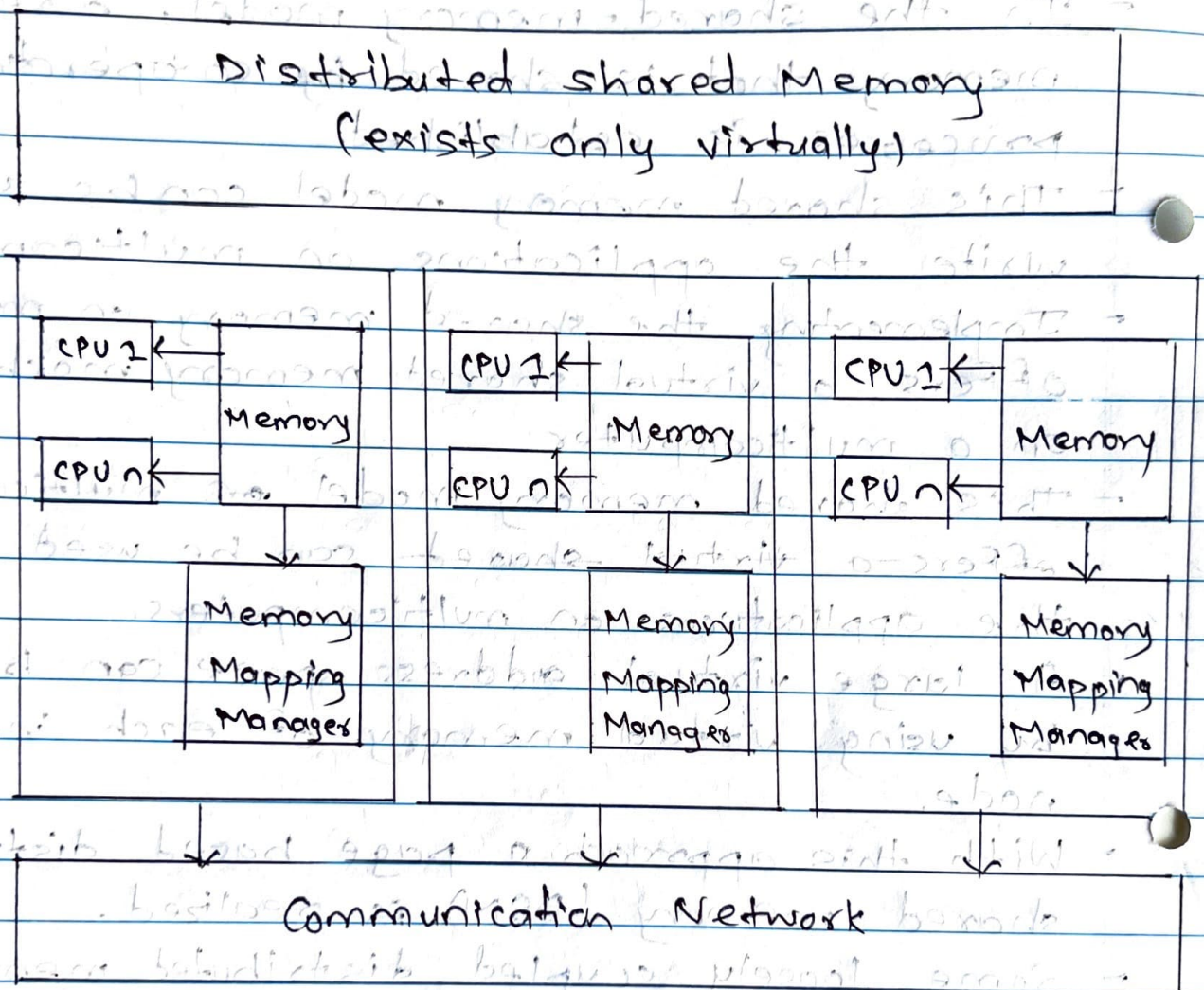
Aim : Distributed Shared Memory

Theory :

- In the shared-memory model, a region of memory that is shared by co-operating processes is established.
- This shared memory model can be used to write the applications on multicomputers.
- Implementing the shared memory on multicomputers offers a virtual shared memory machine, running on a multicomputer.
- This shared memory model on multicomputers offers a virtual shared can be used to write the applications on multicomputers.
- A large virtual address space can be created by using virtual memory of each individual node.
- With this approach a page based distributed shared memory (DSM) is realized.
- Some loosely coupled distributed memory systems have implemented a software layer on the top of message passing system in order to offer shared memory abstraction to the programmers.
- This offers the systems the illusion of physically shared memory and permit programmers to use shared memory paradigms.

• Architecture of DSM:

- Following figure shows the architecture of DSM.



- Every node in the network have one or more CPU and its local memory unit.
- These nodes are connected through high speed communication network.
- A's message passing system is present, it allows the processes running on different nodes in network to exchange messages.

- Each node contains a software memory mapping manager routine which maps local memory of its node to a virtual shared memory space.
- For this mapping purpose the shared memory space is divided into blocks.
- Data caching is supported to reduce network latency where each node's local memory is used to cache the data from virtual address space.
- The memory mapping manager of each node treats local memory as a big cache of the shared memory space for its associated CPU.
- The basic unit of caching is memory block.
- Whenever process on node access the data from memory block of shared address space, then memory mapping manager of that node handle this request.
- If block data in local memory then request will be fulfilled from local memory.
- Otherwise, network fault is generated and control is passed to operating system.
- OS then sends message to the node having that block.
- This missing block is then gets migrated from remote node to requesting process node.
- The memory mapping manager maps this block in application's address space.

- In this way data blocks keep migrating between different nodes.

• Advantages :

- 1) It is easier to implement than RPC.
- 2) DSM improves performance and efficiency by speeding up access to data.
- 3) They all share the address space so one process can easily be moved to a different machine.

• Disadvantages :

- 1) The data access is slow in DSM as compare to non-distributed.
- 2) When programming is done in DSM systems, programmers need to maintain consistency.
- 3) CPU gets slowed down, even cache memory does not aid the situation.

(A)

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